Introduction to Robotics and Engineering Design

SCIENCE
· ON THE COURT ·

A Talk by Dorian Bohler
Physicist and Founder

Session 1

Objective(s):

- Complete student survey
- Robotics Chat
- Navigate to Spike Prime Website

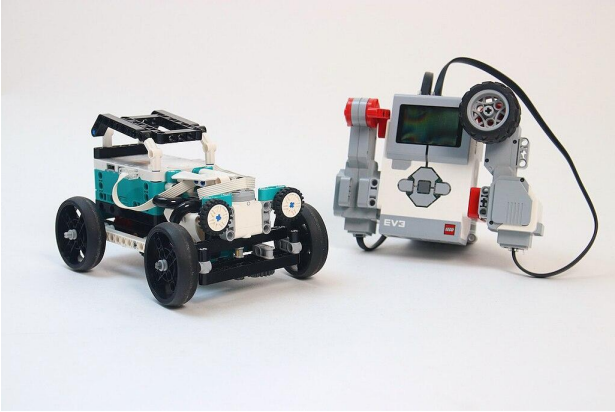
Warm UP

0) Winners of last math challenge

1) What is a robot?

2) How are robots controlled?

3) What makes up a robot?



What is a Robot?

A Robot is a man made machine that is created to solve problems or complete tasks.

Robots feature arms and other lifting mechanism as robotics allows us to surpass human strength

Add caption & link to more info

d by Erection
TOKYO2020
OKYO 2020

What is a Robot?

Robots have brains! Not pink fleshy blobs but supercomputers that allow us to talk to the robots.
Robots also use sensors to find out information. Humans have sensors too! (Eyes, ears, taste buds) and many other ways to find out information on our surroundings.

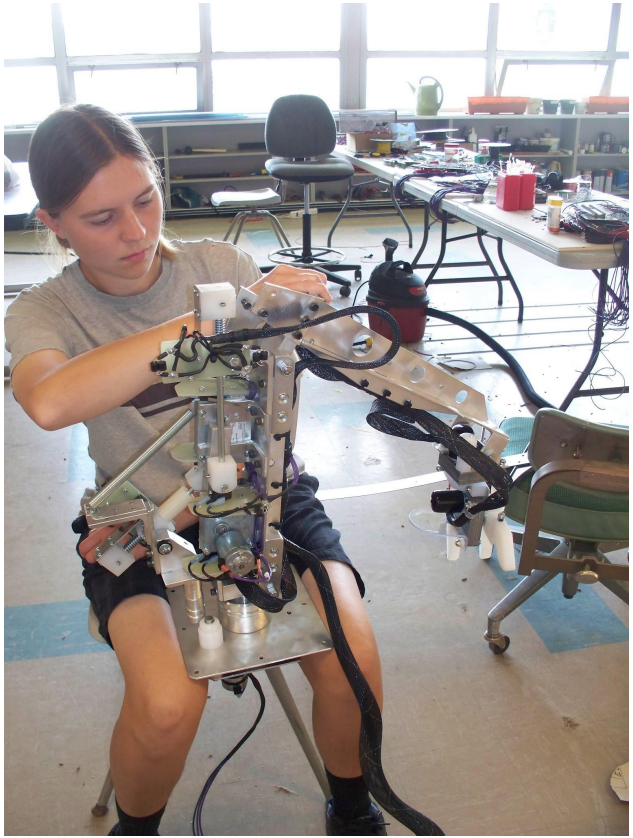
Add caption & link to more info

What is a Robot?

Robots also have wheels! Wheels allow robots and other machines to move across different surfaces.

But how do we move the wheels? Humans communicate with programming to move motors and collect responses from sensors that change how a robot behaves.

Robots are impressive tools only limited by our imaginations!



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Student Survey

https://bit.ly/SOTCsurvey1

📊 MATH CHALLENGE #1 (20 min)

Materials: Rulers, blank paper, $XX prize envelope

Instructions: "Measure your wheel diameter. Calculate circumference. That marker is 40 feet away.

How many wheel rotations to get there? Write your prediction.

Closest wins $XX."

Teacher job: Help with π × diameter formula Prize: Hand out envelope at end, take photo of winner

Explore the brand new LEGO® Education SPIKE™ Prime Set

Taking care of LEGO kits

Please keep all pieces in the appropriate trays

Don't take pieces from other kits

If you drop any pieces on the floor, make sure to pick them up as soon as possible

Spike Prime Website

Show students how to get into the spike prime website

https://spike.legoeducation.com/

Practice the Tutorials

-Turn on motors

-Use the color sensor

-Matrix Array LEDs

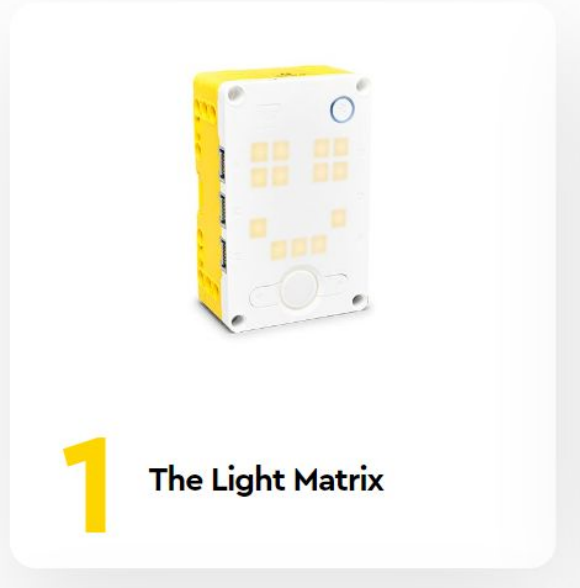
-Use the speaker

How do we engage returning students?

BRAIN

The brain of the robot is where all information is received and where directions are sent from

There are multiple ports built into the brain, these allow for motors and sensors to be connected



How to Test Our Programs

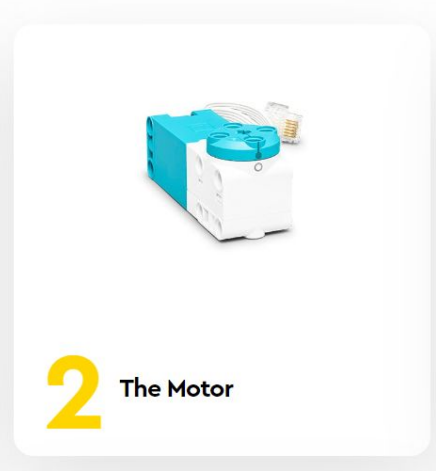
1) Add a wait block to give us time to position our robots
2) Press the button to download the code
3) Press the button to download and run the program

2-4 Seconds

MOTORS

Each moving part needs a motor to enable movement

Rotational motors will spin either clockwise or counterclockwise



SENSORS

Sensors allow the robot brain to process information from the outside world

Color sensor can help the robot identify different colors

Force sensor can detects the robot's collisions

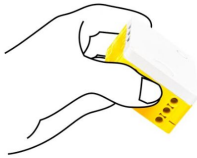
Distance sensor can recognize how far/close something is to the robot

Gyro sensor can identify rotations of the robot

3 The Color Sensor

4 The Distance Sensor

5 The Force Sensor



6 The Gyro Sensor

Python Code w/ Example snippets for each hardware type

How to submit assignments

r in Google Drive

g to add your code for your

beled as: SOTC_FirstName

er

Get G an email address/setup class

Coach G:
Setup google classroom

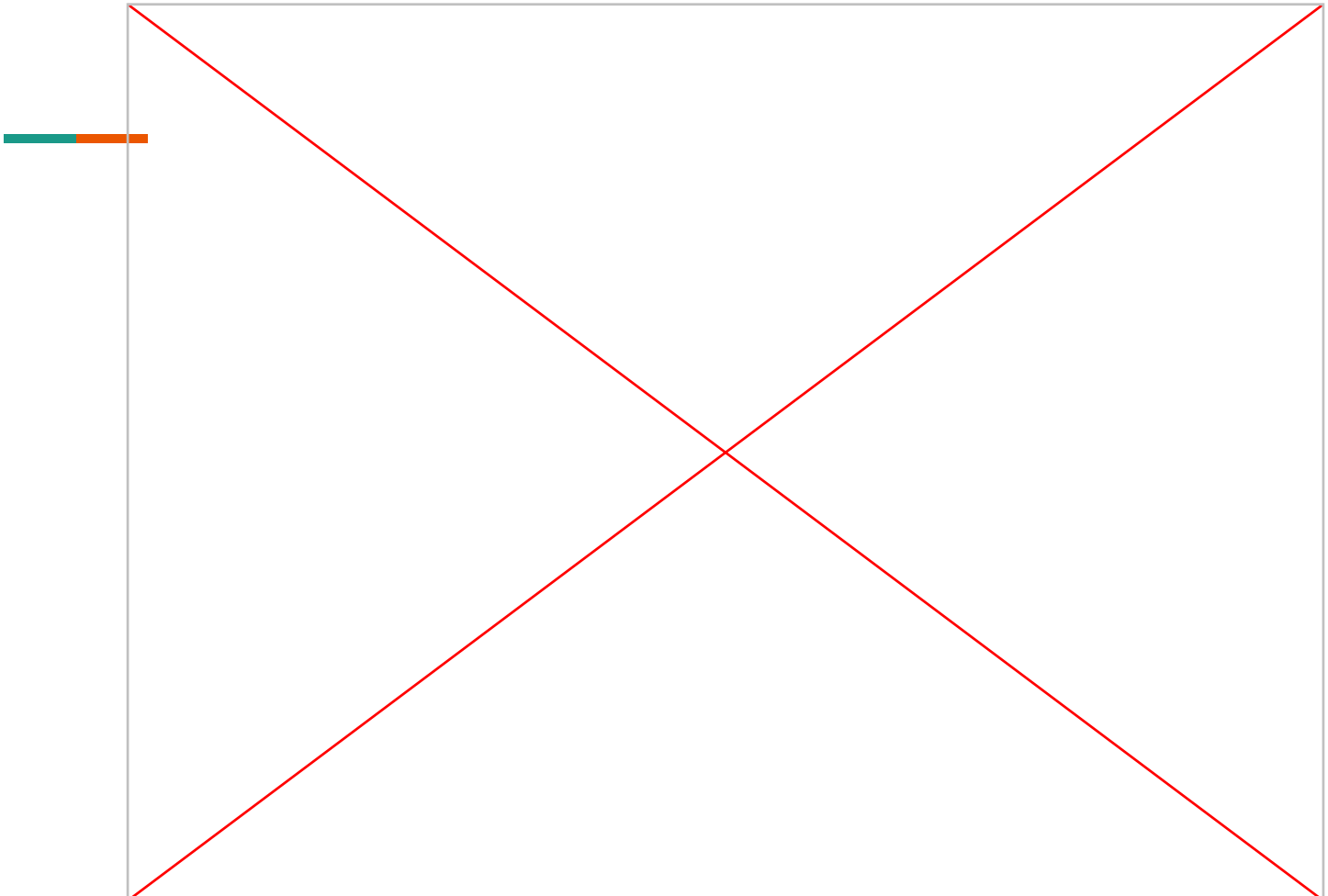
How will assignments be submitted

Session 2

Objective(s):

- Build Driving Base 1
- Check For Understanding
- Introduce Process Flow Diagrams

Navigate to Build Section



Check for Understanding

1) Why does our Robot have 2 Motors instead of 1?
2) Is the size of our wheels important?
3) How did you work with your partner to build the robot?

Build a car than can move forward and backward on the same line?

Challenge #1

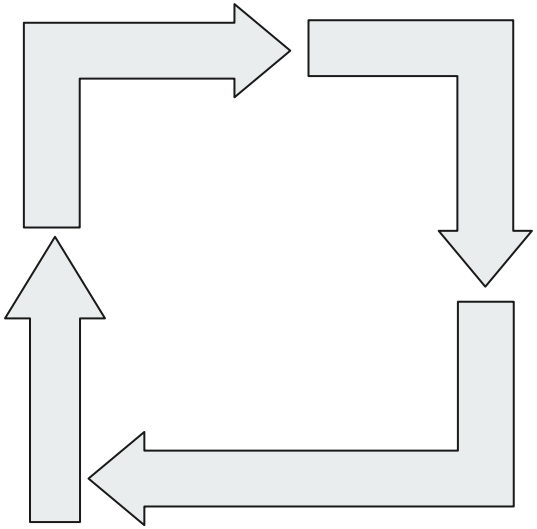
Create a Car that drives straight for X Feet then reverses and stops at the original starting point

Drag Race
Timed Element
Average Time

Word Block and Python

Challenge #2

Create a Robotic car that can follow the path of the arrows show on the right.



Start

Build Course Outside of Classroom

Session 3

Objective(s):

- Check For Understanding
- Introduce the concept of tank/pivot turns
- Complete Challenges 3 and 4

Review Sample Python Code

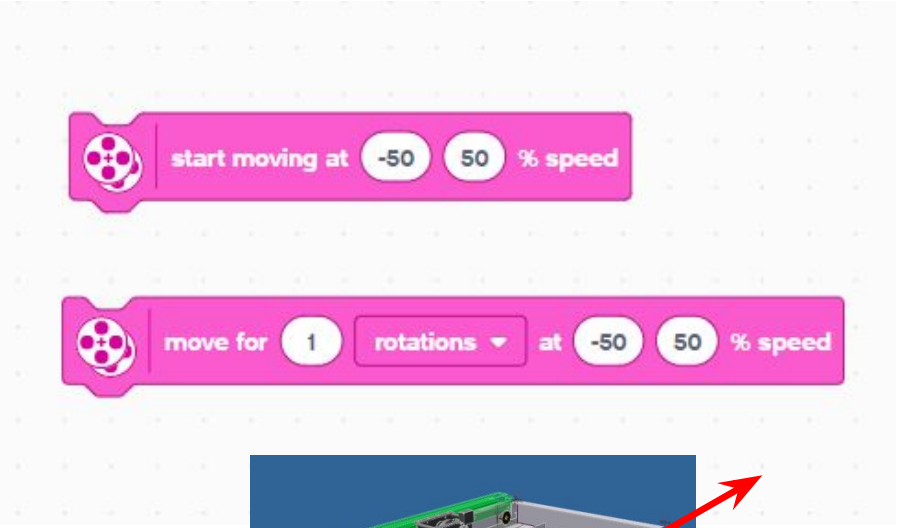
- Instructor provides sample code
- Everyone downloads to cars
- Class gets opportunity to speak to different portions of code

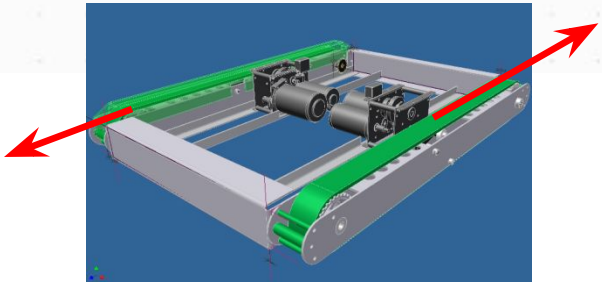
Tank Turn

Tank drive is a special way of steering used by big vehicles, robots, and wheelchairs. It helps them turn precisely and accurately. In tank drive, each side of the robot can move independently.

For example, if we want the robot to go straight, both sides move in the same direction. If we want the robot to turn, one side moves forward while the other side either moves backward.

Tank drive gives us a lot of control and makes steering very accurate, unlike how cars and trucks steer.





TURN IN PLACE

Pivot Turn

When a robot or a person does a pivot turn, they rotate their body around one foot or wheel while keeping the other foot or wheel in the same place.

It helps the robot or person change direction quickly without moving forward or backward.

Just like basketball players use a pivot to spin and change direction while keeping one foot planted, a robot can do the same thing to turn without moving forward or backward.

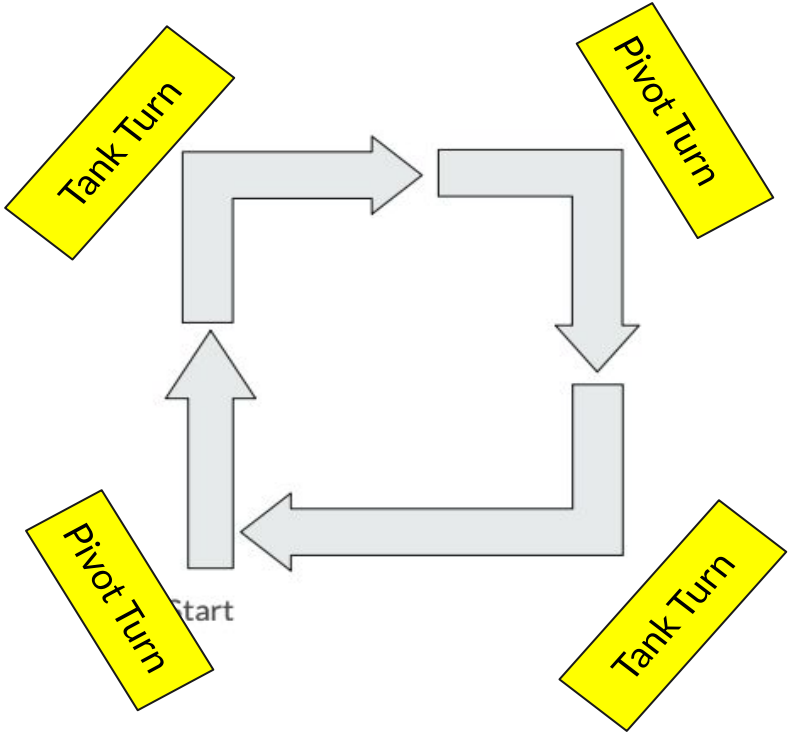
Test Challenge 3 - Using both Types of Turns

1. Get your robot to :
    a. Turn your robot 90 degrees
    b. Turn your robot 180 degrees
    c. Turn your robot 270 degrees
2. Show your instructor Each When Complete

Challenge 4

1. **Create Process Flow Diagram to get your robot to travel in a square using both types of turns!**
2. **Write the code**
3. **Make the robot complete the challenge forward and in reverse**

Add variety to this one



Session 4

Objective(s):

- Check For Understanding
- Circumference vs. Diameter
- Challenge #5 Maze

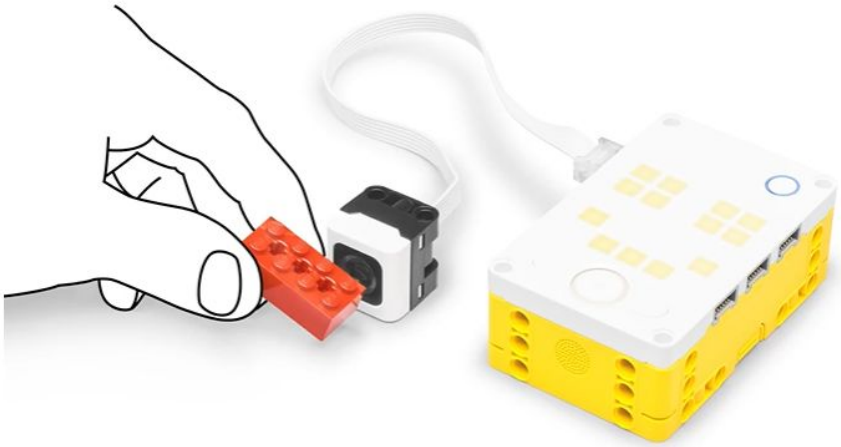
Check for Understanding

1) What types of attachments do robots use?

2)Are robots smart enough to make decisions?

3)How can we teach robots to make decisions?

Color Sensor



Programming the Color Sensor

Students will go through the color sensor learn module to understand the basics!

All sensors work with conditions. This means that when an action happens there will be an reaction.

Action: Seeing a Color

Reaction: Student Task

Color sensor code design

Distance Sensor

The distance sensor allows us to see how close or far we are from objects

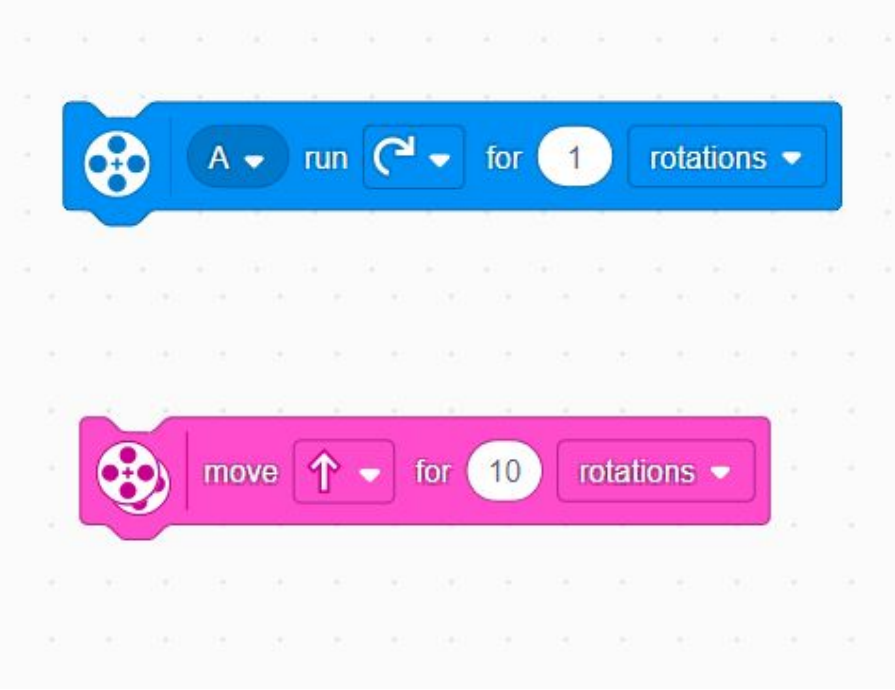
We can use this to trigger certain actions with our robots

See if you can use the color and distance sensors at the same time

Rotations

Students will after measuring set the number of rotations their robot must travel in order to move.

This number has to be calculated....but how?

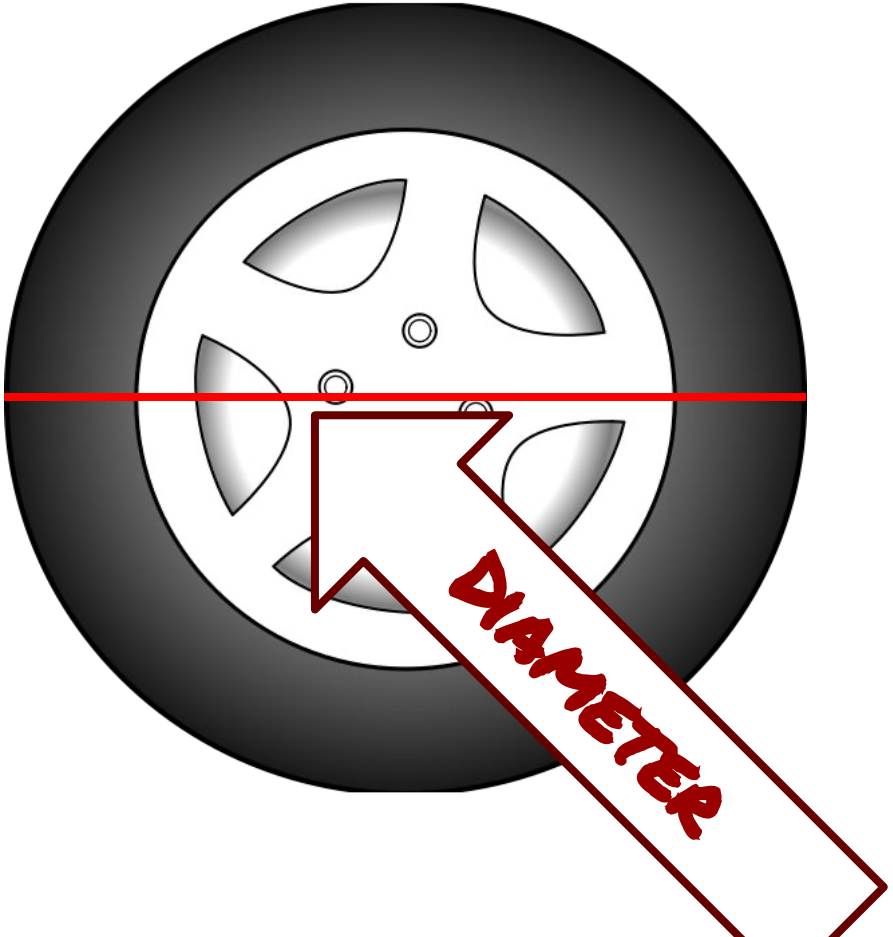


Diameter Explained

Radius is the distance measured with a line from the center of a circle/wheel to its edge

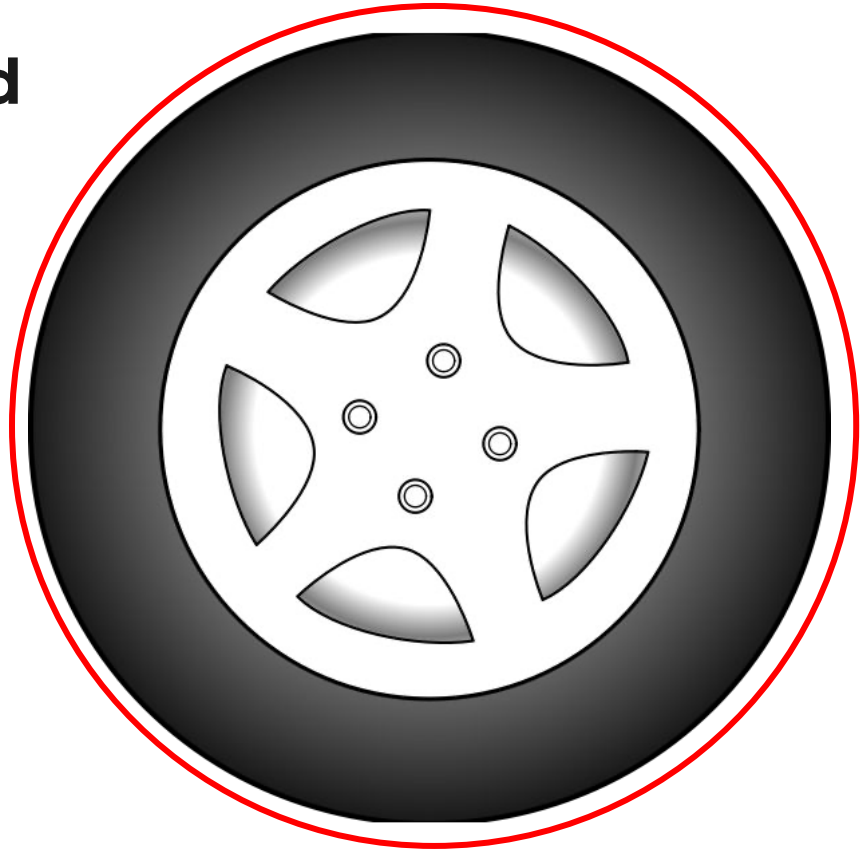
Diameter is the distance from edge to edge thru the center

Diameter = 2 * Radius



Circumference Explained

Circumference is the distance around the edge of a circle



Measured Movement

Step 1:

Find wheel diameter

Step 2:

FInd Circumference of wheel

Step 3:

Use your ruler to find distance between 2 points on the maze

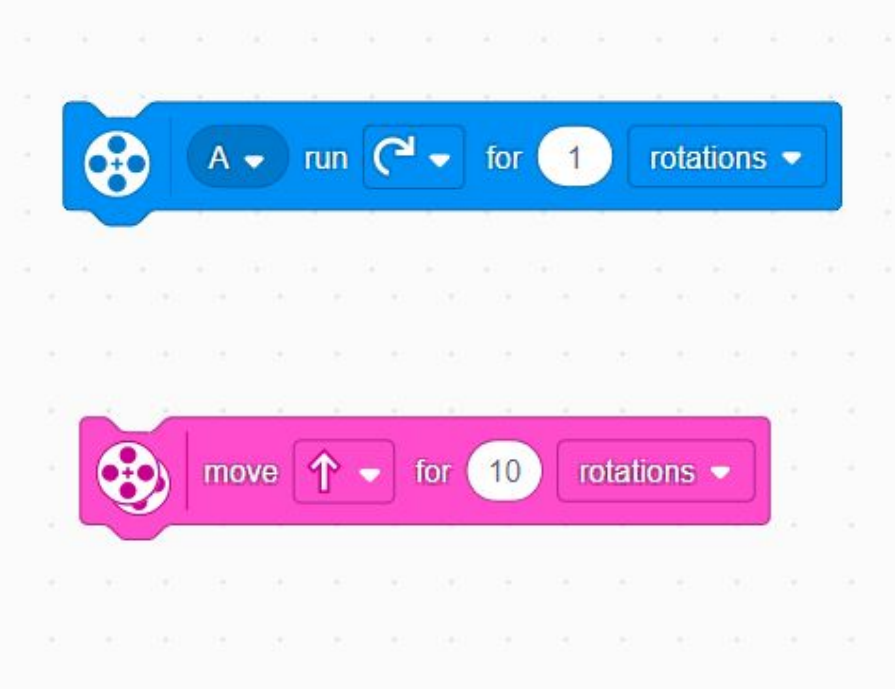
Step 4:

Divide this distance by the circumference. This is how you find the number of rotations.

Rotations

Lets Practice using our newly acquired skill!

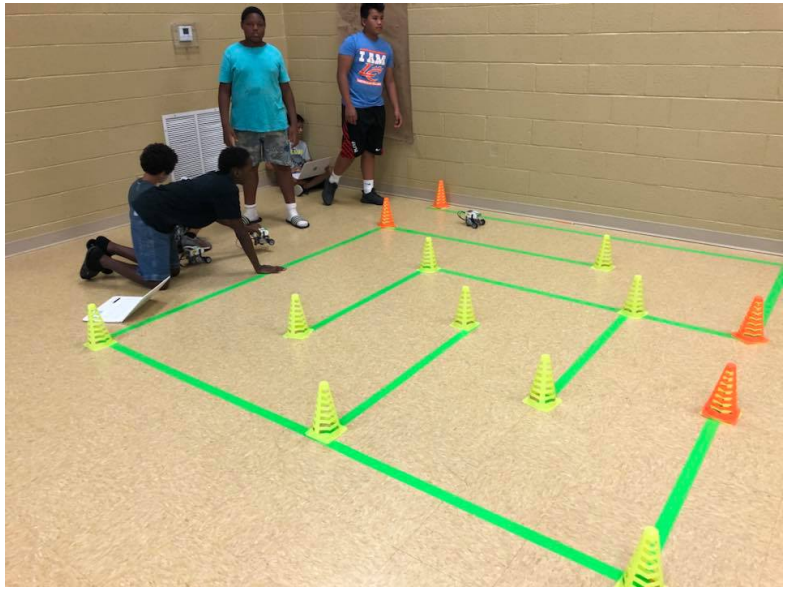
For the next challenge you will measure how many rotations your robot needs to travel in order to complete the activity



Navigating The Maze

From this point on for all projects students must:

- Take Measurements
- Calculate Distances
- Create Process Flow Diagrams
- Document all findings in their Engineering notebooks.
- Create a new program for each activity / challenge



Session 5

Add Instructor answers to previous slide

Add process flow diagram AND code snippet for color usage

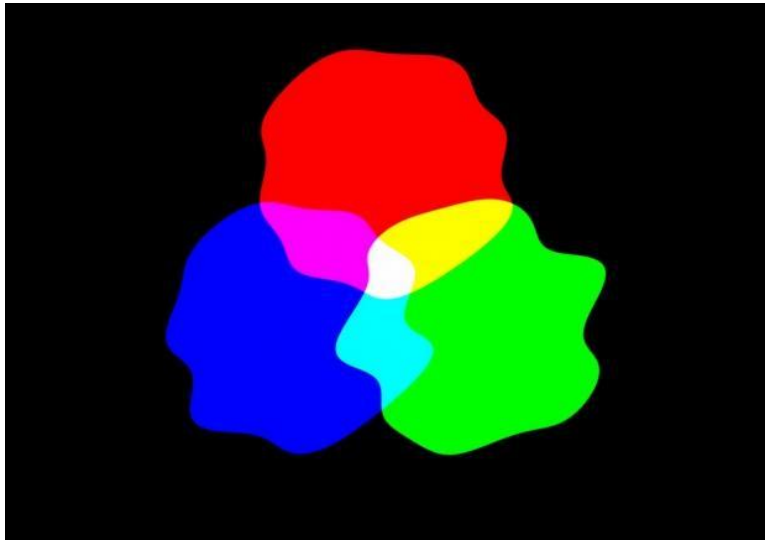
Color Response

Students will be navigating throughout a new maze and he robot will be tasked to do different activities when it see specific colors.

If the robot sees blue, the students will program their robots to say "hello"

If the robot sees red, the students will program their robots to say "Hooray"

If the robot sees green, the students will program their robots to bark like a dog.



Target and Arm Construction

Students will build the arm and use it to collect a target.

This target will be one of 3 colors and the students will be assigned a color by the instructor.

The targets will be scrambled and students must try to collect the most of their targets

Session 6

Decorate Your Cover

Relay Competition

Teams of students collaborate with another team to compete against another grouped team.

The first robot delivers a target to a designated zone.

The second robot retrieves the target, follows the line using the color sensor, and delivers it to the finish zone.

Teams compete to complete the relay first.

References

https://www.youtube.com/watch?v=h2FZKjTsIds

https://spike.legoeducation.com/prime/start/

Robotics and Hoops Challenge

- Team Signups and Roles
- Tournament Guide

Survey for Boys and Girls Club - https://bit.ly/surv0115

Math Assessment(s):

6th Grade - https://bit.ly/six0115

7th Grade - https://bit.ly/sev0115

8th Grade - https://bit.ly/8th0115

Data Science Stuff

New Slide: https://bit.ly/lessonplan212

LinkLink

Robotics Curriculum

10
sessions

Intro to Robotics

Explore Spike Prime kits, learn basic programming concepts, and understand the engineering and design process

- Kit exploration and basic builds
- Block based coding intro
- Simple Design Challenges
- Apply design process to robotics

16
sessions

Robotics & Math

Integrate essential math concepts, like geometry, ratios, data analysis into robotics projects.

- Navigate mazes using geometry
- Experiment with gear ratios
- Collect and analyze sensor data
- Optimize performance

9
sessions

Advanced Design

Tackle complex design challenges, practice iterative testing, and maintain comprehensive documentation.

- Multi-step robotics challenges
- Repeated testing and refinement
- Detailed design documentation
- Reflect on engr. process

14
sessions

Real World & Sports Applications

Apply robotics skills to solve meaningful problems related to sustainability and ethical technology.

- Design Eco-friendly robots
- Explore societal impact of technology
- Develop original projects
- Focus on responsible innovation

5
sessions

Final Project

Finalize designs prepare presentations, and showcase work to peers, mentors, and the community.

- Complete culminating projects
- Create presentation materials
- Practice public speaking
- Host showcase event